

e. Hybrid WAN (SD-WAN) : Introduction to SD-WAN, Key Benefits of SD-WAN

Task 1.

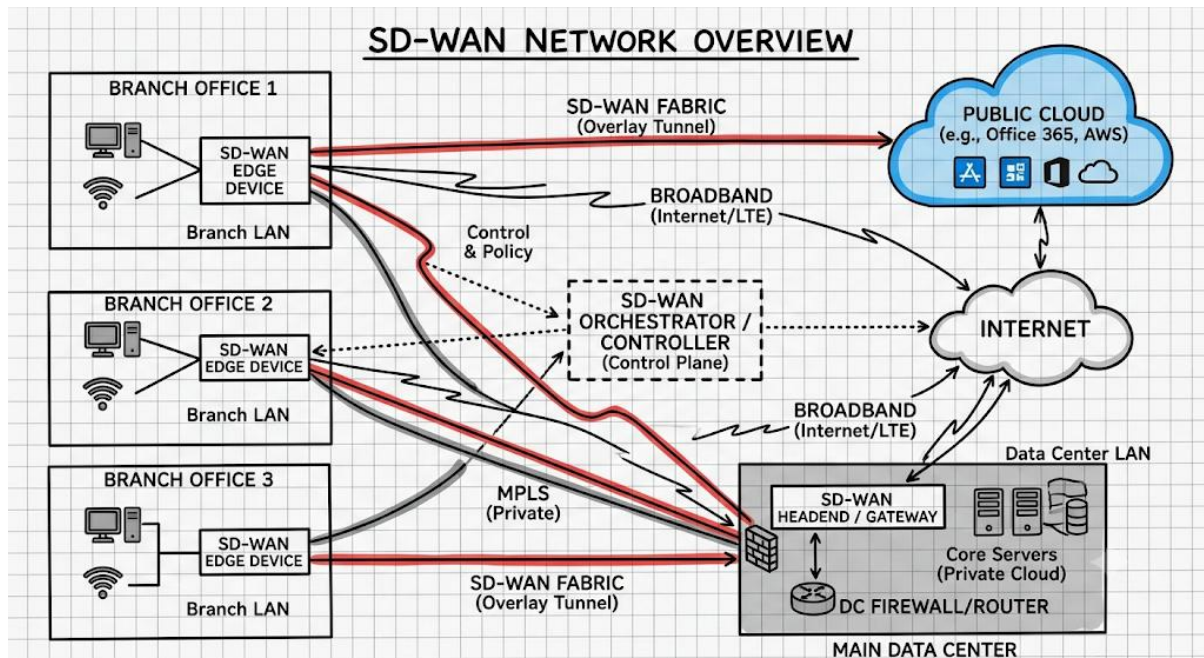
Create a comparison table listing at least 5 key differences between traditional WAN and SD-WAN, focusing on how data is routed, managed, and secured.

Feature	Traditional WAN	SD-WAN
1. Data Routing	Traffic usually follows predefined routes, often through MPLS circuits.	Traffic can dynamically choose the best available path based on performance.
2. Management	Configuration is often performed manually on individual routers.	Centrally managed through an SD-WAN controller/orchestrator.
3. WAN Links	Primarily relies on dedicated links such as MPLS.	Can combine MPLS, broadband, 4G/5G and other connections.
4. Security	Security may depend on separate firewalls and security appliances at each location.	Security policies can be centrally managed and integrated with encryption/firewall capabilities.
5. Application Awareness	Routing is generally based on IP addresses and network destinations.	Can identify applications and prioritize important traffic such as Microsoft 365 or video conferencing.
6. Cost	Dedicated WAN connections can be expensive.	Can use lower-cost broadband/Internet links along with MPLS, potentially reducing WAN costs.
7. Reliability	Failure of the primary WAN link may cause service disruption unless backup links are configured.	Can automatically move traffic to another available link when the preferred path has poor performance or fails.

Task 2.

Draw a simple network diagram (on paper or using a tool like draw.io) showing how SD-WAN connects multiple branch offices to cloud apps (like Office 365 or AWS) and the

main data center. Label the SD-WAN controllers, edges, and at least two types of WAN links (e.g., MPLS, broadband).



Task 3.

Pick any app you use daily (like WhatsApp, Zomato, or Instagram) and write a short scenario explaining how SD-WAN could improve user experience for that app when used across multiple cities in India.

 Hint: Think about speed, reliability, and security when users access the app from different locations.

Example: WhatsApp

- Suppose a company has employees in Ahmedabad, Mumbai, and Bangalore who regularly use WhatsApp for communication with customers and colleagues.
- With traditional WAN, traffic might use a fixed WAN path even when that connection becomes congested or experiences high latency.
- With SD-WAN, the system can monitor available WAN links such as MPLS and broadband and select an appropriate path based on network conditions.
- For example, if the broadband connection in Mumbai becomes slow, SD-WAN can move important WhatsApp-related traffic to another available link.
- Traffic can also be protected using encrypted SD-WAN tunnels and security policies.

Result:

- **Better speed:** Traffic can use a suitable available path.
- **Higher reliability:** Traffic can move to another link when a connection fails.
- **Better security:** Traffic between company locations can be protected using secure tunnels and centralized security policies.
- **Better user experience:** Employees can communicate with customers with fewer connectivity disruptions.

Task 4.

List and explain three key benefits of SD-WAN for a company with offices in Ahmedabad, Mumbai, and Bangalore. For each benefit, give a real-world example of how it would help employees or customers.

1. Better Reliability

SD-WAN can use multiple WAN connections and dynamically select an available path.

Example:

An employee in the Ahmedabad office is attending an online meeting. If the primary MPLS connection develops a problem, SD-WAN can use the broadband connection instead, helping maintain connectivity.

2. Better Performance and Application Prioritization

SD-WAN can recognize different types of application traffic and apply policies to important applications.

Example:

Employees in Mumbai are using Microsoft Teams for an important meeting while other employees are downloading large files. SD-WAN can prioritize the real-time meeting traffic to help reduce latency and disruption.

3. Centralized Management and Security

SD-WAN allows network administrators to manage policies from a central platform instead of configuring every branch router individually.

Example:

A company opens a new branch in Bangalore. The administrator can apply the

company's WAN and security policies centrally to the new SD-WAN edge, making deployment and ongoing management easier.